

Assignment: MCQ Questions based on Cars24 Case Study

Answer Key with Code Explanations

Q1 Answer:

```
df = pd.read_csv('train-cars24-car-price.csv') # Load CSV file into DataFrame  
df.head() # Display first 5 rows
```

Q2 Answer:

```
df.shape # Returns tuple (rows, columns)
```

Q3 Answer:

```
df.info() # Summarizes data types, non-null count, memory usage
```

Q4 Answer:

```
df.describe() # Statistical summary for numerical columns
```

Q5 Answer:

```
df['fuel_type'].value_counts() # Count of each unique value
```

Q6 Answer:

```
df[df['mileage'] == 0] # Filter rows where mileage is zero
```

Q7 Answer:

```
mean_mileage = df[df['mileage'] > 0]['mileage'].mean() # Compute mean excluding zeros  
df['mileage'] = df['mileage'].replace(0, mean_mileage) # Replace 0s
```

Q8 Answer:

```
df['make'] = df['full_name'].apply(lambda x: x.split()[0]) # Extract brand
```

Q9 Answer:

```
df.drop('full_name', axis=1, inplace=True) # Drop column
```

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Q10 Answer:

```
sns.boxplot(x='fuel_type', y='selling_price', data=df) # Boxplot  
  
plt.title('Selling Price by Fuel Type')  
  
plt.show()
```

Q11 Answer:

```
sns.heatmap(df.corr(), annot=True, cmap='YlGnBu') # Correlation heatmap  
  
plt.title('Feature Correlation')  
  
plt.show()
```

Q12 Answer:

```
trans_dummies = pd.get_dummies(df['transmission_type'], drop_first=True) # One-hot encode  
  
df = pd.concat([df.drop('transmission_type', axis=1), trans_dummies], axis=1)
```

Q13 Answer:

```
df.sort_values('selling_price', ascending=False).head(10) # Top 10 expensive cars
```

Q14 Answer:

```
for t in df['seller_type'].unique():  
    sns.kdeplot(df[df['seller_type'] == t]['selling_price'], label=t)  
  
plt.legend()  
  
plt.title('Selling Price KDE by Seller Type')  
  
plt.show()
```

Q15 Answer:

```
df['age'] = 2020 - df['year']  
  
plt.scatter(df['age'], df['selling_price'])  
  
plt.title('Age vs Selling Price')
```

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```
plt.xlabel('Age')
```

```
plt.ylabel('Selling Price')
```

```
plt.show()
```

Q16 Answer:

```
df.loc[df['selling_price'] > 100, 'selling_price'] = 100 # Cap at 100
```

Q17 Answer:

```
df['make'].unique() # List of unique brands
```

Q18 Answer:

```
percentage = (df[df['mileage'] > 40].shape[0] / df.shape[0]) * 100
```

```
print(f'{percentage:.2f}% of cars have mileage > 40')
```

Q19 Answer:

```
df['log_selling_price'] = np.log(df['selling_price']) # Natural log
```

Q20 Answer:

```
df.sample(3) # Randomly select 3 rows
```